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Lubricant pump system with rapid coupling reservoir

Abstract

A lubricant pump system includes a pump assembly which has a housing with an interior pumping chamber having an inlet. At least one pumping unit is disposed at least partially within the pumping chamber and is configured to displace lubricant from the pumping chamber through an outlet. The housing has a pump interface with an opening fluidly coupleable with the pumping chamber inlet and a pump valve controlling lubricant flow through the pump interface opening. A reservoir includes a body connectable with the pump housing and having a central axis, an interior storage chamber for containing a quantity of lubricant and a reservoir interface engageable with the pump interface. The reservoir interface includes an opening fluidly coupleable with the storage chamber and fluidly coupleable with the pump interface opening. The reservoir body includes a reservoir valve controlling lubricant flow through the reservoir interface opening.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention relates to lubrication systems, and more particularly to pumps for lubrication systems.

(2) Lubrication systems for providing lubricant to mechanical components such as bearings, gear trains, etc. typically include one or more pumps to direct lubricant from a reservoir to such mechanical components. Such pumps may either supply the lubricant to an intermediate delivery device, for example an injector, or supplies lubricant directly to the mechanical component by hoses or fluid lines. In certain applications, the reservoir is directly attached to a pump assembly and the reservoir must be periodically refilled. Typically, such refilling of the reservoir requires delivering the lubricant into the pump reservoir from an external resupply reservoir, generally by means of a hose attached to a fill port, such that a certain amount of maintenance time is required for such refilling operations.

SUMMARY OF THE INVENTION

(3) In one aspect, the present invention is a lubricant pump system comprising a pump assembly including a housing with a central axis and an interior pumping chamber having an inlet. At least one pumping unit is disposed at least partially within the pumping chamber and is configured to displace lubricant from the pumping chamber through an outlet. The housing has a pump interface with an opening fluidly coupleable with the pumping chamber inlet and a pump valve controlling lubricant flow through the pump interface opening. A reservoir includes a body connectable with the pump housing and having a central axis, an interior storage chamber for containing a quantity of lubricant and a reservoir interface engageable with the pump interface. The reservoir interface includes an opening fluidly coupleable with the storage chamber and fluidly coupleable with the pump interface opening. Further, the reservoir body includes a reservoir valve controlling lubricant flow through the reservoir interface opening.

(4) In another aspect, the present invention is again a lubricant pump system comprising a pump assembly including a housing with a central axis and an interior pumping chamber having an inlet. At least one pumping unit is disposed at least partially within the pumping chamber and is configured to displace lubricant from the pumping chamber through an outlet. The housing has a pump interface with an opening fluidly coupled with the pumping chamber inlet and a pump valve controlling lubricant flow through the pump interface opening and including a fixed valve seat and a movable valve plug. A reservoir includes a body connectable with the pump housing and having a central axis, an interior storage chamber for containing a quantity of lubricant and a reservoir interface engageable with the pump interface. The reservoir interface includes an opening fluidly coupled with the storage chamber and fluidly coupleable with the pump interface opening. Further, the reservoir body has a reservoir valve controlling lubricant flow through the reservoir interface

opening and including a fixed valve seat and a movable valve plug. The reservoir valve plug contacts the pump valve seat and the reservoir valve seat contacts the pump valve plug when the reservoir interface engages with the pump interface. As such, relative displacement of the reservoir toward the pump assembly displaces both the reservoir valve plug and the pump valve plug generally along the central axis and fluidly couples the reservoir storage chamber with the pumping chamber.

(5) In a further aspect, the present invention is yet again a lubricant pump system comprising a pump assembly including a housing with a central axis and an interior pumping chamber having an inlet. At least one pumping unit is disposed at least partially within the pumping chamber and is configured to displace lubricant from the pumping chamber through an outlet. The housing includes a main body portion providing the pumping chamber and an adapter block attached to the main body portion and an outer axial end providing a pump interface. The pump interface has an opening fluidly coupled with the pumping chamber inlet. A reservoir includes a body having a central axis, a tubular portion providing an interior storage chamber for containing a quantity of lubricant, and an adapter block releasably connectable with the pump adapter block to connect the reservoir with the pump assembly. The pump adapter block is attached to the tubular portion and has an outer axial end providing a reservoir interface engageable with the pump interface. Further, the reservoir interface has an opening fluidly coupled with the storage chamber and fluidly coupled with the pump interface opening when the reservoir interface is engaged with the pump interface, so as to fluidly connect the reservoir storage chamber with the pumping chamber.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) The foregoing summary, as well as the detailed description of the preferred embodiments of the present invention, will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the invention, there is shown in the drawings, which are diagrammatic, embodiments that are presently preferred. It should be understood, however, that the present invention is not limited to the precise arrangements and instrumentalities shown. In the drawings:

- (2) FIG. 1 is an elevational view of the lubricant pump system of the present invention;
- (3) FIG. 2 is a schematic view of a lubrication system including the lubricant pump system;
- (4) FIG. 3 is an axial cross-sectional view in perspective of the lubricant pump system;
- (5) FIG. 4 is a front elevational view of the lubricant pump, showing a reservoir separate from a pump assembly;
- (6) FIG. 5 is a perspective view from the top of the pump assembly;
- (7) FIG. 6 is a perspective view from the bottom of the reservoir;
- (8) FIG. 7 is a broken-away, enlarged axial cross-sectional view of the lubricant pump system, showing a reservoir interface spaced from a pump interface;
- (9) FIG. 8 is a broken-away, enlarged axial cross-sectional view of the lubricant pump, showing an initial engagement of the reservoir interface with the pump interface;
- (10) FIG. 9 is a broken-away, enlarged axial cross-sectional view of the lubricant pump, showing a full engagement of the reservoir interface with the pump interface;
- (11) FIG. 10 is a more enlarged view of a portion of FIG. 9, showing a flow passage through a reservoir valve and pump valve;
- (12) FIG. 11 is a broken-away, enlarged axial cross-sectional view of the pump assembly, showing an exemplary pumping unit;
- (13) FIG. 12 is a broken-away perspective view of the lubricant pump system, showing a plurality of connectors being positioned for engagement;

- (14) FIG. 13 is an enlarged view of a portion of FIG. 12, showing the details of the presently preferred connectors;
- (15) FIG. 14 is an axial cross-sectional view of a pump interface/adaptor block;
- (16) FIG. 15 is a top plan view of a reservoir interface/adaptor block;
- (17) FIG. 16 is an axial cross-sectional view of the reservoir interface/adaptor block; and
- (18) FIG. 17 is an axial cross-sectional view of a reservoir valve seat.

DETAILED DESCRIPTION OF THE INVENTION

(19) Certain terminology is used in the following description for convenience only and is not limiting. The words “inner”, “inwardly” and “outer”, “outwardly” refer to directions toward and away from, respectively, a designated centerline or a geometric center of an element being described, the particular meaning being readily apparent from the context of the description. Further, as used herein, the words “connected” and “coupled” are each intended to include direct connections between two members without any other members interposed therebetween and indirect connections between members in which one or more other members are interposed therebetween. The terminology includes the words specifically mentioned above, derivatives thereof, and words of similar import.

(20) Referring now to the drawings in detail, wherein like numbers are used to indicate like elements throughout, there is shown in FIGS. 1-17 a lubricant pump system 10 for supplying lubricant, such as grease, within a lubrication system 1. The lubrication system 1 may also include one or more lubricant injectors 2 for delivering lubricant to a mechanical device 3, such a bearing 4, a gear train (not shown), etc., such that the pump system 10 directs lubricant to such injector(s) 2, as best shown in FIG. 2. Alternatively or additionally, the pump assembly 10 may directly supply the lubricant to the mechanical device(s) 3 through one or more hoses or fluid lines 5 and divider valves 6, as also shown in FIG. 2. The lubricant pump system 10 basically comprises a pump assembly 12 and a lubricant reservoir 14, and preferably includes a pump interface block 16 and a reservoir interface block 18, as described in detail below.

(21) More specifically, the pump assembly 12 includes a housing 20, the housing 20 having a central axis A.sub.CP, an interior pumping chamber 22 with an inlet 24, and at least one and preferably a plurality of pumping units 26 (FIG. 11). The one or more pumping units 26 are each disposed at least partially within the pumping chamber 22 and are configured to displace lubricant from the pumping chamber 22 through an outlet 28, as described in further detail below. The housing 20 has a pump interface 30 with an opening 32 fluidly coupleable with the pumping chamber inlet 24 and preferably includes a pump valve 34 controlling lubricant flow through the pump interface opening 32.

(22) Further, the reservoir 14 basically includes a body 40 connectable with the pump housing 20 and having a central axis A.sub.CR, which is coaxial with the pump central axis A.sub.CP when the reservoir 14 is connected with the pump assembly 12, an interior storage chamber 42 for containing a quantity of lubricant and a reservoir interface 44 engageable with the pump interface 30. The reservoir interface 44 includes an opening 46 fluidly coupleable with the storage chamber 42 and fluidly coupleable with the pump interface opening 32. Also, as with the pump housing 20, the reservoir body 40 preferably includes a reservoir valve 48 controlling lubricant flow through the reservoir interface opening 46.

(23) With the basic structure above, the pump valve 34 seals the pump interface opening 32 and the reservoir valve 48 seals the reservoir interface opening 46 when the reservoir 14 is separate from the pump assembly 12. As such, lubricant is retained within the reservoir 14 by the reservoir valve 48 and the pump valve 34 both prevents any lubricant from leaking from the pump housing 20, or perhaps more significantly, substantially prevents air from entering into the pumping chamber 22. Further, the reservoir valve 48 interacts with the pump valve 34 when the reservoir interface 44 engages with the pump interface 30 so as to open both the pump interface opening 32 and the reservoir interface opening 46, thereby fluidly connecting the reservoir storage chamber 42 with the

pumping chamber **22**, as discussed below, for subsequent delivery to the injector(s) **2** or directly to the mechanical device(s) **3**.

(24) Furthermore, the pump assembly **12** or/and the reservoir **14** includes at least one connector **50** (FIGS. **12** and **13**) configured to releasably connect the reservoir **14** with the pump assembly **12**, thereby maintaining engagement of the reservoir and pump interfaces **44**, **30**. Preferably, each one of the interface blocks **16**, **18** has components of a bayonet connection, as described in detail below, to releasably connect the reservoir **14** with the pump assembly **12**. However, the reservoir **14** and the pump assembly **12** may be releasably connected by any other appropriate means, such as for example, by one or more clamps (none shown) attached to the pump assembly **12** and engageable with the reservoir **14**, etc.

(25) Referring to FIGS. **7-10**, the pump valve **34** preferably includes a fixed valve seat **35** and a movable valve plug **37**. The valve plug **37** is engageable with the pump valve seat **35** so as to close the pump interface opening **32** and is displaceable or “space-able” from the seat **35** to allow lubricant flow through the opening **32**. Similarly, the reservoir valve **48** includes a fixed valve seat **47** and a movable valve plug **49** engageable with the pump valve seat **47**. As such, the reservoir interface opening **46** is closed or sealed when the plug **49** engages the seat **47** and is open when the plug **49** is spaced or displaced from the seat **47**.

(26) With the above-described structure of the two valves **34**, **48**, the reservoir valve plug **49** contacts the pump valve seat **35** and the reservoir valve seat **47** contacts the pump valve plug **37** when the reservoir interface **44** engages with the pump interface **30**. As such, further relative displacement of the reservoir **14** toward the pump assembly **12** displaces both the reservoir valve plug **49** (i.e., from the seat **47**) and the pump valve plug **37** (from seat **35**) generally along the central axes A.sub.CP, A.sub.CR and fluidly couples the reservoir chamber **42** with the pumping chamber **22**, i.e., through the fluidly connected interface openings **32**, **46**. Specifically, an annular passage **17** is defined between the reservoir valve seat **47** and the pump valve seat **35** when the reservoir **14** is connected with the pump assembly **12**, as best shown in FIG. **10**, the passage **17** fluidly coupling the reservoir storage chamber **42** with the pumping chamber **22**.

(27) Referring to FIG. **7**, each one of the pump valve **34** and the reservoir valve **48** preferably further includes at least one and preferably a plurality of biasing members **60**, **62**, respectively, (only one each shown) configured to bias each respective valve plug **37**, **49** to the closed positions shown in FIGS. **4-8**. Specifically, the one of more biasing members **60** are disposed within the pump housing **20** and are configured to bias the pump valve plug **37** axially against the pump valve seat **35**, so as to thereby prevent lubricant flow through the pump valve seat **35** (i.e., through interface opening **30**) when the pump assembly **12** is separate from the reservoir **14**. Likewise, the biasing member(s) **62** of the reservoir valve **48** are configured to bias the reservoir valve plug **49** axially against the reservoir valve seat **47**, thereby preventing lubricant flow through the valve seat **49**/interface opening **46** when the reservoir **14** is separate from the pump assembly **12**. Thus, both valves **34**, **48** are normally closed when each of the primary system components **12**, **14** are separate from each other, but preferably interact with each other to fluidly connect the interface openings **32**, **46**, and thus fluidly couple the reservoir storage chamber **42** with the pumping chamber **22**, simultaneously with engagement of the pump and reservoir interfaces **30**, **44**.

(28) Consequently, the lubricant system **10** of the present invention enables a substantially immediate resupply of lubricant to the pump assembly **12** merely by pushing the reservoir interface **44** into engagement with the pump interface **30**, and the lubricant pump system **10** is thereafter fully operational simply by securing the reservoir **14** to the pump assembly **12** as discussed above and in further detail below. Having described the basic structure and functions above, these and other elements of the present lubricant pump system **10** are described in further detail below.

(29) Referring now to FIGS. **3**, **6-9** and **11**, the pump housing **20** preferably includes a main body portion **70** and the pump interface block **16**, as mentioned above. The main body portion **70** provides the pumping chamber **22** and contains or houses the one or more pumping units **26**.

Preferably, the main body portion **70** is complex-shaped and includes a lower cavity **71**, an upper cavity **72** providing the pumping chamber **22** and an upper circular flow plate **73** attached to an upper end **70a** of the main body portion **70**. The flow plate **73** has a central hub **74**, an outer rim **75** and a plurality of radial flanges **76** connecting the rim **75** to the hub **74** and defining a plurality of flow ports **77** collectively providing the pump inlet **24**, as best shown in FIG. 3.

(30) In a presently preferred embodiment, the pump assembly **12** includes a motor **78** disposed within the lower cavity **71** of the main body portion **70**, along with related drive components (none indicated), and has a shaft **78a** extending along the pump central axis A.sub.CP. An eccentric drive member or “cam” **79** is disposed on the motor shaft **78a** and within the pumping chamber **22**, and at least one and preferably a plurality of the pumping units **26** are arranged about the drive member **79**. Referring particularly to FIG. 11, each pumping unit **26** preferably includes a cylindrical sleeve **80** extending through an opening **81** in the housing main body portion **70**, a piston **82** disposed within the sleeve **80** and a spring **83**. Specifically, each sleeve **80** has a central passage **84**, an inlet **80a** disposed within the pumping chamber **22** and an outlet **80b** providing one pump outlet **28**, which is connectable with a hose or other fluid line **4**, preferably by means of a coupler **85**. Each piston **82** is slidably disposed within the sleeve central passage **84** and has an outer end **82a** biased by the spring **83** into engagement with the eccentric drive member **79**.

(31) As such, rotation of the drive member **79** reciprocates the piston **82** to alternatively draw lubricant into the central passage **84** when displaced toward the drive member **79** by the spring **83**, and then to direct lubricant within the passage **84** through the sleeve outlet **80b** when displaced or pushed by the drive member **79** toward the outlet **80b**. Although the plurality of pumping units **26** as described above are presently preferred, the pump assembly **12** may include at least one pumping unit **26** formed in any other appropriate manner, such as for example a centrifugal pump or another type of positive displacement pump. The lubricant pump system **10** of the present invention is in no manner limited to any specific structure of the pumping unit(s) **26** or to any other pumping elements for directing and/or pressurizing lubricant.

(32) Referring now to FIGS. 7-9 and 14, the pump interface block **16** preferably includes a circular cylindrical body **86** having an outer axial end **86a** providing the pump interface **30**, an inner axial end **86b** attached to the upper end **70a** of the housing main body portion **70**, and an outer circumferential surface **87**. The body **86** of the interface block **16** has an interior cavity **88** and an inner opening **89** through the body inner axial end **86b**. The inner opening **89** is fluidly coupled with the pump inlet **24**, to thereby fluidly connect the interface block interior cavity **88** with the pumping chamber **22**. Also, the interface block body **86** further has an inner circumferential surface **90** extending axially inwardly from the outer axial end **86a** to the interior cavity **88**. The inner circumferential surface **90** is preferably frustoconical, defines the pump interface opening **32** and provides the pump valve seat **35**.

(33) Preferably, the body **86** of the pump interface block **16** further has a central annular projection **91** extending outwardly from the outer axial end **80a** and providing the valve seat **35**, and thus also the inner circumferential surface **90**. The projection **91** also has an outer circumferential surface **92** disposeable within the reservoir block **18**, as discussed below, and a flat outer annular surface **94**. Further, the reservoir block body **86** preferably has a counterbore section **96** extending inwardly from the inner axial end **80b** and sized to receive a portion of the upper end **70a** of the pump housing main body portion **70**, as well as the flow plate **73** mounted thereto, so as to removably connect the pump interface block **16** with the main body portion **70** of the pump housing **20**. As such, the pump interface block **16** preferably functions as an adapter block and is configured to connect with a conventional, most preferably commercially available, pump. However, the interface block **16** may alternatively be fabricated as an integral component of the pump assembly **12** and connected or integrally formed with the remainder of the pump housing **20** by any appropriate means.

(34) Referring to FIGS. 12 and 13, the pump interface block body **86** preferably includes at least

one and preferably a plurality of L-shaped retainer tabs **98** projecting outwardly from and spaced circumferentially about the body outer axial end **86a** adjacent to the body outer circumferential surface **87**. Each retainer tab **98** is disposeable within a mating L-shaped recess **152** of the reservoir body **40** and has a wedge-shaped arm **98a** engageable with the body **40** to provide one bayonet connector **50**, as described above and in further detail below. Further, an annular rim **99** extends outwardly from the adapter block body outer end **86a** so as to generally enclose the plurality of tabs **98**, the rim **99** providing a section of the body outer circumferential surface **87**.

(35) Referring to FIGS. 7-10, the valve plug **37** of the pump valve **34** is movably disposed at least partially within the interior cavity **88** of the pump interface block **16**. The plug **37** is configured to prevent flow from the interior cavity **88**, or externally of the block **16** into the cavity **88**, through the block outer axial end **80a** when engaged with the valve seat **35**. Preferably, the pump valve plug **37** includes a circular plate **100** with opposing axial ends **100a**, **100b** and is disposeable within the pump interface opening **32**. The circular plate **100** has an outer circumferential surface **102** sealingly engageable with the valve seat inner circumferential surface **90** and a flat circular surface **104** on the outer axial end **100a**. Further, the plug outer circumferential surface **102** preferably has an outer frustoconical surface section **102a** which is disposable against the inner frustoconical surface **90** of the valve seat **35** to seal the pump interface opening **32**, i.e., when urged by the biasing members **60**. When the valve plug **37** is engaged with the valve seat **35**, the valve plug circular surface **104** is substantially flush with the valve seat flat surface **94**, i.e., the surfaces **94** and **104** are disposed within a common plane VP.sub.p, as indicated in FIG. 7.

(36) Preferably, the pump assembly **12** further includes a circular retainer plate **106** with a plurality of pockets **107** each containing a portion of one biasing member **60**, which are each preferably formed as a compression coil spring. The retainer plate **106** includes a plurality of mounting tabs **108** connectable by fasteners (not shown) to an inner radial surface **88a** partially defining the interior cavity **88**. The pump assembly **12** also preferably includes a stirrer **110** disposed within the interior cavity **88** of the interface block **16** for circulating lubricant contained therein to thereby release any trapped air. The stirrer **110** preferably includes a central hub **112** attached to an outer end of the motor shaft **78a** and a stirring paddle **114** extending radially from the hub **112**, such that rotation of the motor shaft **78a**, in addition to driving the cam **79**, rotates the hub **112** to angularly displace the stirrer paddle **114** within the interior cavity **88**.

(37) Referring to FIGS. 1, 3 and 6, the reservoir body **40** preferably includes a tubular portion **120**, which provides the lubricant storage chamber **42**, and the reservoir interface block **18** as mentioned above. The body tubular portion **120** has a circular cylindrical sidewall **122** with opposing first and second axial ends **122a**, **122b** and an end cap **124** enclosing the first axial end **122a**. As best shown in FIG. 3, the reservoir **14** includes a circular cylindrical follower **132** disposed within the storage chamber **42** and configured to rest upon a surface (not shown) of lubricant contained within the chamber **42**. A guide rod **134** extends axially through the chamber **42** between the end cap **124** and the reservoir interface block **18** and through a central opening **133** of the follower **132**. Further, the reservoir tubular portion **120** preferably has a threaded inner surface section **123** formed in the sidewall **122** and configured to threadably engage with a mating outer threaded surface section **142a** of the reservoir interface block **18**, as described below.

(38) Referring to FIGS. 7-9, 15 and 16, the reservoir interface block **18** preferably includes a generally circular cylindrical body **140** having an outer axial end **140a** providing the reservoir interface **44**, an opposing inner axial end **140b**, an outer circumferential surface **142**, and an interior cavity **144** fluidly coupled with the storage chamber **42**. Preferably, the outer surface **142** has the threaded section **142a** threadably engaged with the threaded inner surface section **123** of the reservoir tubular portion **120**, as discussed above. Thus, the tubular portion **120** and reservoir interface block **18** are releasably connected, such that the interface block **18** functions as an adapter block connectable to a conventional or commercially available reservoir. However, the interface block **18** and the tubular portion **120** of the reservoir body **40** may be connected in any other

appropriate manner, either releasably or permanently, and/or specially fabricated as components of an integral design.

(39) Further, the cylindrical body **140** of the reservoir block **18** preferably has an inner circumferential surface **145** extending inwardly from the body outer axial end **140a** and at least partially defining the reservoir cavity **144**. Adjacent to the body inner axial end **140b**, the interface cylindrical body **140** preferably includes a central hub portion **146**, an outer rim portion **147** and a plurality of radial spokes **148** spaced circumferentially about the reservoir central axis A.sub.CR and defining a plurality of generally triangular passages **149**. Each passage **149** extends from the body inner axial end **140b** to the interior cavity **144** and fluidly connect the reservoir storage chamber **42** with the cavity **144**. Furthermore, a cylindrical projection **150** extends outwardly from the central hub portion **146** into the storage chamber **42** and receives an end **134a** of the follower guide rod **134**.

(40) Referring again to FIGS. **12** and **13**, as mentioned above, the reservoir interface block **18** preferably includes at least one and most preferably a plurality of recesses **152** with a generally triangular or wedge-shaped engagement section **152a** spaced circumferentially about the reservoir central axis A.sub.CR. Each recess **152** extends axially from the body outer axial end **140a** and radially inwardly from the outer circumferential surface **142**, with the engagement section **152a** extending circumferentially from a remainder of the recess **152** so as to define a triangular/wedge-shaped retainer section **153**. Further, each recess **152** is configured to receive a separate one of the L-shaped tabs **98** of the pump interface block **16** when the reservoir **14** is displaced axially into engagement with the pump assembly **12**.

(41) Specifically, the reservoir block outer axial end **140a** becomes disposed within the pump block annular rim **99** and each tab **98** enters a separate recess **152** when the reservoir interface **44** engages with the pump interface **30**, to open the valves **34**, **48**, and then the reservoir **14** is angularly displaced or turned about the central axis A.sub.C through a limited angular displacement or fractional turn. Such angular displacement/turning engages the arm **98a** of each tab **98** of the pump interface block **16** with the retainer section **153** formed by a separate recess **152** of the reservoir interface block **18** to releasably connect the reservoir **14** with the pump assembly **12** without requiring the use of any tools. However, the reservoir **14** may be releasably coupled with the pump assembly **12** to maintain engagement of the pump and reservoir interfaces **30**, **44**, respectively, by any other appropriate means, such as for example, one or more clamps, etc.

(42) Referring now to FIGS. **7-10** and **17**, the reservoir valve plug **49** is movably disposed at least partially within the interior cavity **144** of the reservoir interface block **18**. The plug **49** is configured to prevent flow from the interior cavity **144** through the interface block outer axial end **140a** when engaged with the reservoir valve seat **47**. Preferably, the valve plug **49** includes an annular body **154** with an inner circumferential surface **155**, an outer circumferential surface **156**, and opposing inner and outer circular/annular end surfaces **157A**, **157B**, respectively. The outer circumferential surface **155** is sealingly engaged with the inner circumferential surface **145** of the interior cavity **144** and the inner circumferential surface **155** defines the reservoir interface opening **46** and is sealingly engageable with the valve seat **49**. Specifically, the inner surface **155** has a frustoconical surface section **155a** engageable with the valve seat **47** as described below. Further, at least the outer axial end surface **157B** is a substantially flat and is contactable by the flat circular surface **94** of the pump valve seat **35**, for reasons discussed below.

(43) As best shown in FIG. **7**, the reservoir interface block **18** preferably includes an annular spring retainer **160** disposed within the interior cavity **144** and having a plurality of pockets **162** (one shown) each retaining a portion of one of the biasing members **62**, which are each preferably formed as a compression coil spring. The spring retainer **160** is disposed against the radial spokes **148** of the interface block cylindrical body **140** and retained axially by the force exerted by the biasing members **62**.

(44) Referring again to FIGS. **7-10** and **17**, the reservoir valve seat **47** preferably includes a

cylindrical body **164** disposed within the interior cavity **144** of the reservoir interface block **18**. The cylindrical body **164** of the valve seat **47** has an outer circumferential surface **166**, an inner axial end **164a** disposed against the central hub portion **146** of the cylindrical body **140**, and an outer axial end **160b** with a flat circular surface **168** generally flush with the outer axial end **140a** of the body **140** of the reservoir interface block **18**. Preferably, the outer circumferential surface **166** has a frustoconical surface section **166a** contactable by, and sealingly engageable with, the frustoconical inner surface section **155a** of the valve plug **49** to seal the reservoir interface opening **46**. With the engaging inner and outer frustoconical surfaces **155a**, **166a**, the valve seat **47** prevents axial displacement of the valve plug **49** out of the interior cavity **144** of the reservoir interface block **18**. When the valve plug **49** is engaged with the valve seat **47**, the valve plug annular surface **157B** is substantially flush with the valve seat flat circular surface **168**, i.e., the surfaces **157B** and **168** are disposed within a common plane VP.sub.R, as indicated in FIG. 7. Also, the flat outer circular surface **168** of the reservoir valve seat **47** is configured to contact the flat circular surface **104** of the pump valve plug **37** during engagement of the reservoir and pump interfaces **30**, **44**, as discussed below.

(45) Referring to FIGS. 4, 7-10, 12 and 13, with the above-described preferred structures of the pump assembly **12** and the reservoir **14**, particularly the respective interface blocks **16**, **18** thereof, the present lubricant pump **10** is rapidly and conveniently refilled with lubricant generally in the following manner. When the quantity of lubricant contained within the reservoir storage chamber **42** has been depleted, the empty reservoir **14** is first uninstalled from the pump assembly **12**. Preferably, the reservoir **14** is removed from the pump assembly **12** by a limited angular displacement or fractional turn of the reservoir **14**, such that each retainer tab **98** of the pump assembly **12** is withdrawn from the associated retainer recess **152** of the reservoir **14**. As the reservoir interface block **18** uncouples from the pump interface block **16**, the pump valve **34** closes the pump interface opening **32** and the reservoir valve **48** simultaneously closes the reservoir interface opening **46**, so as thereby prevent any residual lubricant from escaping from the pump assembly **12** or from the reservoir **14**, and also prevents air from entering into each interface block **16**, **18**.

(46) Then, another reservoir **14** with a filled storage chamber **42** is conveyed to the pump assembly **12** and positioned such that the reservoir interface **44** is adjacent to the pump interface **30**, as shown in FIG. 7. The reservoir **14** is then displaced axially toward the pump assembly **12** (i.e., along the coaxial central axes A.sub.CR, A.sub.CP) such that the reservoir valve plug **49** contacts the pump valve seat **35** and the reservoir valve seat **47** contacts the pump valve plug **37**, as depicted in FIG. 8. Due to the contacting flat surfaces **94**, **157B** of the pump valve seat **35** and the reservoir valve plug **49**, respectively, and the contacting flat surfaces **104**, **168** of the pump valve plug **37** and the reservoir valve seat **47**, respectively, air is pushed out of the space between the reservoir **14** and the pump assembly **12** when the reservoir interface **44** is displaced into engagement with the pump interface **30**.

(47) As the reservoir interface **44** is further displaced axially into engagement with the pump interface **30**, the pump fixed valve seat **35** displaces the reservoir movable valve plug **49** axially into the interior cavity **144** of the reservoir interface block **18** as the pump valve seat **35** also becomes disposed within the reservoir cavity **144**, as shown in FIGS. 9 and 10. Simultaneously, the reservoir fixed valve seat **47** displaces the pump movable valve plug **37** axially into the interior cavity **88** of the pump interface block **16** as the reservoir valve seat **47** enters the pump cavity **88** and a portion of the valve seat **47** becomes disposed within the pump valve seat **35**. As such, the annular passage **17** becomes defined between the frustoconical inner circumferential surface **90** of the pump valve seat **35** and the frustoconical outer circumferential surface section **166a** of the reservoir valve seat **47**, thereby fluidly coupling the interior cavity **144** of the reservoir interface block **18** and the interior cavity **88** of the pump interface block **16**, and thus the reservoir storage chamber **42** with the pumping chamber **22**. Finally, the reservoir **14** is turned/rotated about the

reservoir central axis A.sub.CR to engage the pump retainer tabs **98** with the reservoir recesses **152** and thereby releasably connect the reservoir **14** with the pump assembly **12**.

(48) Representative, non-limiting examples of the present invention were described above in detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the invention.

(49) Moreover, combinations of features and steps disclosed in the above detailed description may not be necessary to practice the invention in the broadest sense, and are instead taught merely to particularly describe representative examples of the invention. Furthermore, various features of the above-described representative examples, as well as the various independent and dependent claims below, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

(50) All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter. The invention is not restricted to the above-described embodiments, and may be varied within the scope of the following claims.

Claims

1. A lubricant pump system comprising: a pump assembly including a housing with a central axis, an interior pumping chamber having an inlet and at least one pumping unit disposed at least partially within the pumping chamber and configured to displace lubricant from the pumping chamber through an outlet, the housing having a pump interface with an opening fluidly coupleable with the pumping chamber inlet and a pump valve controlling lubricant flow through the pump interface opening; and a reservoir including a body connectable with the housing and having a central axis, an interior storage chamber for containing a quantity of lubricant and a reservoir interface engageable with the pump interface, the reservoir interface including an opening fluidly coupleable with the storage chamber and fluidly coupleable with the pump interface opening, the reservoir body including a reservoir valve controlling lubricant flow through the reservoir interface opening; wherein the pump valve includes a fixed valve seat and a movable valve plug engageable with the pump valve seat so as to close the pump interface opening and the reservoir valve includes a fixed valve seat and a movable valve plug engageable with the reservoir valve seat so as to close the reservoir interface opening, the reservoir valve plug contacting the pump valve seat and the reservoir valve seat contacting the pump valve plug when the reservoir interface engages with the pump interface such that relative displacement of the reservoir toward the pump assembly displaces both the reservoir valve plug and the pump valve plug generally along the central axis of the housing and fluidly couples the reservoir chamber with the pumping chamber.
2. The lubricant pump system as recited in claim 1 wherein: the pump valve seals the pump interface opening and the reservoir valve seals the reservoir interface opening when the reservoir is separate from the pump; and the reservoir valve interacts with the pump valve when the reservoir interface engages with the pump interface so as to open the pump interface opening and the reservoir interface opening to fluidly connect the reservoir storage chamber with the pumping chamber.
3. The lubricant pump system as recited in claim 1 wherein at least one of the pump assembly and the reservoir includes at least one connector configured to releasably connect the reservoir with the pump assembly.

4. The lubricant pump system as recited in claim 1 wherein an annular passage is defined between the reservoir valve seat and the pump valve seat when the reservoir body is connected with the housing, the annular passage fluidly coupling the reservoir storage chamber with the pumping chamber.

5. The lubricant pump system as recited in claim 1 wherein: the pump valve further includes at least one biasing member configured to bias the pump valve plug axially against the pump valve seat so as to prevent lubricant flow through the pump valve seat when the pump assembly is separate from the reservoir; and the reservoir valve further includes at least one biasing member configured to bias the reservoir valve plug axially against the reservoir valve seat so as to prevent lubricant flow through the reservoir valve seat when the reservoir is separate from the pump assembly.

6. The lubricant pump system as recited in claim 1 wherein the pump valve seat has a flat outer annular surface contactable with a flat outer annular surface of the reservoir valve plug and the pump valve plug has a flat circular surface contactable with a flat circular surface of the reservoir valve seat such that air is pushed out of a space between the reservoir and the pump assembly as the reservoir interface displaces into engagement with the pump interface.

7. The lubricant pump system as recited in claim 1 wherein: the housing includes a main body portion, the main body portion providing the pumping chamber and containing the at least one pumping unit, and an interface block attached to the main body portion and having an outer axial end providing the pump interface and an interior cavity fluidly coupled with the pump inlet, the pump valve plug being movably disposed at least partially within the cavity of the pump interface block and configured to prevent flow from the interior cavity through the outer axial end of the pump interface block when engaged with the pump valve seat; and the reservoir body includes a tubular portion providing the lubricant storage chamber and an interface block attached to the tubular portion and having an outer axial end providing the reservoir interface and an interior cavity fluidly coupled with the storage chamber, the reservoir valve plug being movably disposed at least partially within the interior cavity of the reservoir interface block and configured to prevent flow from the interior cavity through the interface block outer axial end when engaged with the reservoir valve seat, the reservoir interface block being releasably attachable to the pump interface block to connect the reservoir with the pump assembly.

8. The lubricant pump system as recited in claim 7 wherein: the pump interface block has an inner circumferential surface extending axially inwardly from the outer axial end of the pump interface block, the inner circumferential surface defining the pump interface opening and providing the pump valve seat, and the pump valve plug includes a circular plate disposeable within the pump interface opening and having an outer circumferential surface sealingly engageable with the valve seat inner circumferential surface; and the reservoir interface block includes an inner circumferential surface at least partially defining the reservoir cavity, the reservoir valve seat includes a cylindrical body disposed within the interior cavity of the reservoir interface block and having an outer circumferential surface, and the reservoir valve plug includes an annular body disposed within the interior cavity of the reservoir interface block and about the reservoir valve seat cylindrical body, the valve plug annular body having an outer circumferential surface sealingly engaged with the inner circumferential surface of the reservoir interface block and an inner circumferential surface defining the reservoir interface opening and being sealingly engageable with the outer circumferential surface of the reservoir valve seat.

9. The lubricant pump system as recited in claim 8 wherein the pump interface block has an annular projection extending from the block outer axial end and providing the valve seat, the inner circumferential surface of the pump interface block being provided on the annular projection, the annular projection being disposeable within the reservoir block cavity when the reservoir interface engages with the pump interface.

10. The lubricant pump system as recited in claim 8 wherein a portion of the reservoir valve seat is

disposed within the pump valve seat when the reservoir interface is engaged with the pump interface such that an annular passage is defined between the outer circumferential surface of the reservoir valve seat and the inner circumferential surface of the pump valve seat, the annular passage fluidly coupling the interior cavity of the reservoir interface block and the interior cavity of the pump interface block.

11. The lubricant pump system as recited in claim 7 wherein: the pump valve further includes at least one biasing member disposed within the cavity of the pump interface block and configured to bias the pump valve plug axially toward the outer axial end of the pump interface block and against the pump valve seat so as to prevent lubricant flow through the pump interface opening when the pump assembly is separate from the reservoir; and the reservoir valve further includes at least one biasing member disposed within the cavity of the reservoir interface block and configured to bias the reservoir valve plug axially toward the outer axial end of the reservoir interface block and against the reservoir valve seat so as to prevent lubricant flow through the reservoir interface opening when the reservoir is separate from the pump assembly.

12. The lubricant pump system as recited in claim 7 wherein the at least one pumping unit includes a motor with a shaft, an eccentric drive member disposed on the motor shaft and within the pumping chamber, a cylindrical sleeve having a central passage, an inlet disposed within the pumping chamber and an outlet providing the pump outlet, and a piston slidably disposed within the sleeve central passage and engageable by the eccentric drive member such that rotation of the drive member reciprocates the piston within the sleeve so as to direct lubricant entering the sleeve inlet through the sleeve outlet.

13. The lubricant pump system as recited in claim 12 wherein the pump further includes a stirrer attached to the motor shaft and disposed within the interior cavity of the pump interface block such that rotation of the motor shaft angularly displaces the stirrer within the interior cavity.

14. A lubricant pump system comprising: a pump assembly including a housing with a central axis, an interior pumping chamber having an inlet and at least one pumping unit disposed at least partially within the pumping chamber and configured to displace lubricant from the pumping chamber through an outlet, the housing having a pump interface with an opening fluidly coupleable with the pumping chamber inlet and a pump valve controlling lubricant flow through the pump interface opening and including a fixed valve seat and a movable valve plug; and a reservoir including a body connectable with the housing and having a central axis, an interior storage chamber for containing a quantity of lubricant and a reservoir interface engageable with the pump interface, the reservoir interface including an opening fluidly coupleable with the storage chamber and fluidly coupleable with the pump interface opening, the reservoir body having a reservoir valve controlling lubricant flow through the reservoir interface opening and including a fixed valve seat and a movable valve plug, the reservoir valve plug contacting the pump valve seat and the reservoir valve seat contacting the pump valve plug when the reservoir interface engages with the pump interface such that relative displacement of the reservoir toward the pump assembly displaces both the reservoir valve plug and the pump valve plug generally along the central axis of the housing and fluidly couples the reservoir storage chamber with the pumping chamber.

15. The lubricant pump system as recited in claim 14 wherein: the pump valve seals the pump interface opening and the reservoir valve seals the reservoir interface opening when the reservoir is separate from the pump; and the reservoir valve interacts with the pump valve when the reservoir interface engages with the pump interface so as to open the pump interface opening and the reservoir interface opening to fluidly couple the reservoir chamber with the pump inlet.

16. The lubricant pump as recited in claim 14 further comprising at least one connector configured to releasably connect the reservoir with the pump assembly.

17. The lubricant pump system as recited in claim 14 wherein: the housing includes a main body portion, the main body portion providing the pumping chamber and containing the at least one pumping unit, and an interface block attached to the main body portion and having an outer axial

end providing the pump interface and an interior cavity fluidly coupled with the pump inlet, the pump valve plug being movably disposed at least partially within the cavity of the pump interface block and configured to prevent flow from the interior cavity through the pump interface opening when engaged with the pump valve seat; and the reservoir body includes a tubular portion providing the lubricant storage chamber and an interface block having an inner axial end attached to tubular portion, an outer axial end providing the reservoir interface, and an interior cavity fluidly coupled with the storage chamber, the reservoir valve plug being movably disposed at least partially within the interior cavity of the reservoir interface block and configured to prevent flow from the interior cavity through the reservoir interface opening when engaged with the reservoir valve seat, the reservoir interface block being releasably attachable to the pump interface block to connect the reservoir with the pump assembly.

18. The lubricant pump system as recited in claim 17 wherein: the pump interface block has an inner circumferential surface extending axially inwardly from the outer axial end of the pump interface block, the inner circumferential surface defining the pump interface opening and providing the pump valve seat, and the pump valve plug includes a circular plate disposeable within the pump interface opening and having an outer circumferential surface sealingly engageable with the valve seat inner circumferential surface; and the reservoir interface block includes an inner circumferential surface at least partially defining the reservoir cavity, the reservoir valve seat includes a cylindrical body disposed within the interior cavity of the reservoir interface block and having an outer circumferential surface, and the reservoir valve plug includes an annular body disposed within the interior cavity of the reservoir interface block and about the reservoir valve seat cylindrical body, the valve plug annular body having an outer circumferential surface sealingly engaged with the inner circumferential surface of the reservoir interface block and an inner circumferential surface defining the reservoir interface opening and being sealingly engageable with the outer circumferential surface of the reservoir valve seat.
